Hao Zhu

CURRICULUM VITAE

Ph.D. Neuroscience

Brain and Mind Institute | The Chinese University of Hong Kong

4/F Hui Yeung Shing Building, Shatin, N.T., Hong Kong SAR, China

haozhu@cuhk.edu.hk / hao-zhu.com

Academic Appointments

2024-present **Postdoctoral Fellow**
Brain and Mind Institute, The Chinese University of Hong Kong
Department of Linguistics and Modern Languages
Hong Kong SAR, China
Advisors: Patrick C. M. Wong; Xiangbin (Shawn) Teng

2017-2019 **Research Associate**
NYU-ECNU Institute of Brain and Cognitive Science
Shanghai, China

Education

2019-2024 **Ph.D. in Cognitive Neuroscience**
New York University, New York, NY, USA
Dissertation: *Investigation of Motor-based Predictive Signals in Speech Production.*
Advisor: Xing Tian

2013-2017 **B.Sc. in Neural Science**
New York University / Shanghai, New York, NY, USA
Capstone: *Mental Imagery in Human Speech Production and Perception*

Awards and Honors

2026 Post-Doctoral Fellow Award, CNS 2026

2026 BMI Data Blitz Best Presentation Award, CUHK

2020 Dean's Conference Fund, NYU

2019 MacCracken Doctoral Fellowship, NYU

2019 NYU Shanghai Doctoral Fellowship, NYU Shanghai

Teaching Experience

2021 **Teaching Assistant, EEG Systems Laboratory**
East China Normal University (Spring 2021)

Skills

Technical Python; MATLAB; JavaScript; TypeScript; Node.js; MEG/EEG/iEEG analysis; PCB & 3D printing; Computational modeling

Languages English (fluent); Mandarin Chinese (native)

Publications

* Equal contribution

2026 Li, X., Zheng, X., Wu, Y., **Zhu, H.**, Shu, Y, Tong, R., & Tian, X.
Motor-based prediction during preparation of hand movement modulates auditory processing in two distinct directions. *NeuroImage.*

2024 Yang, F., **Zhu, H.***, Cao, X., Li, H., Fang, X., Yu, L., Li, S., Wu, Z., Li, C., Zhang, C., & Tian, X.
Impaired motor-to-sensory transformation mediates auditory hallucinations. *PLOS Biology.*

2024 Yu, X., Li, J., **Zhu, H.**, Tian, X., & Lau, E.
Electrophysiological hallmarks for event relations and event roles in working memory. *Frontiers in Neuroscience.*

2023 Han, Z., **Zhu, H.**, Shen, Y., & Tian, X.
Segregation and integration of sensory features by flexible temporal characteristics of independent neural representations. *Cerebral Cortex.*

2022 Zheng, X., **Zhu, H.**, Li, S., & Tian, X.
The generic inhibitory function of corollary discharge in motor intention: Evidence from modulation effects of speech preparation on late components of auditory neural responses. *eNeuro.*

2021 Yang, F., **Zhu, H.**, Yu, L., Lu, W., Zhang, C., & Tian, X.
Deficits in multi-scale top-down processes distorting auditory perception in schizophrenia. *Behavioural Brain Research.*

2020 Li, S., **Zhu, H.***, & Tian, X.
Corollary discharge versus efference copy: Distinct neural signals in speech preparation differentially modulate auditory responses. *Cerebral Cortex.*

2018 Yang, J., **Zhu, H.***, & Tian, X.
Group-level multivariate analysis in EasyEEG toolbox: Examining the temporal dynamics using topographic responses. *Frontiers in Neuroscience.*

Presentations

2026 **Zhu, H.** (Presenter). Intracranial mapping of speech entrainment and frequency-following responses.
Society for the Neurobiology of Language (SNL), Geneva, Switzerland.

2026 **Zhu, H.** (Presenter). Auditory Network Integration and Neural Representations Collapse Under General Anesthesia: Evidence from Intracranial EEG and Self-Supervised Learning.
Cognitive Neuroscience Society (CNS), Vancouver, BC, Canada.

2019 **Zhu, H.** (Presenter). Corollary discharge versus efference copy: Distinct neural signals in speech preparation differentially modulate auditory responses.
Society for Neuroscience (SfN), Chicago, IL, USA.

References

Patrick C. M. Wong, Ph.D.
Stanley Ho Professor of Cognitive Neuroscience
Department of Linguistics and Modern Languages
The Chinese University of Hong Kong
Hong Kong SAR, China

Xing Tian, Ph.D.
Associate Professor of Neural and Cognitive Sciences
Department of Neural and Cognitive Sciences
New York University Shanghai
Shanghai, China

Xiangbin Teng, Ph.D.
Assistant Professor of Psychology
Department of Psychology
The Chinese University of Hong Kong
Hong Kong SAR, China